Applied Agentic AI for PMs/TPMs, capstone project

PRD Genie

An AI pipeline that turns meeting notes into PRDs and user stories, without inventing requirements

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Turning meetings into documents is slow, uneven and risky

Rewriting

Transcripts and notes are turned into structured documents by hand.

Inconsistent formats

Every PM structures a PRD differently, so engineering must decode each one.

Lost requirements

What was said in the meeting does not always reach the PRD.

◆ The biggest risk: an AI that invents a requirement nobody stated

Three stages, with the PM approving each one

Analyze

Extracts only what was said, with exact quotes, and asks about what is missing.

PRD

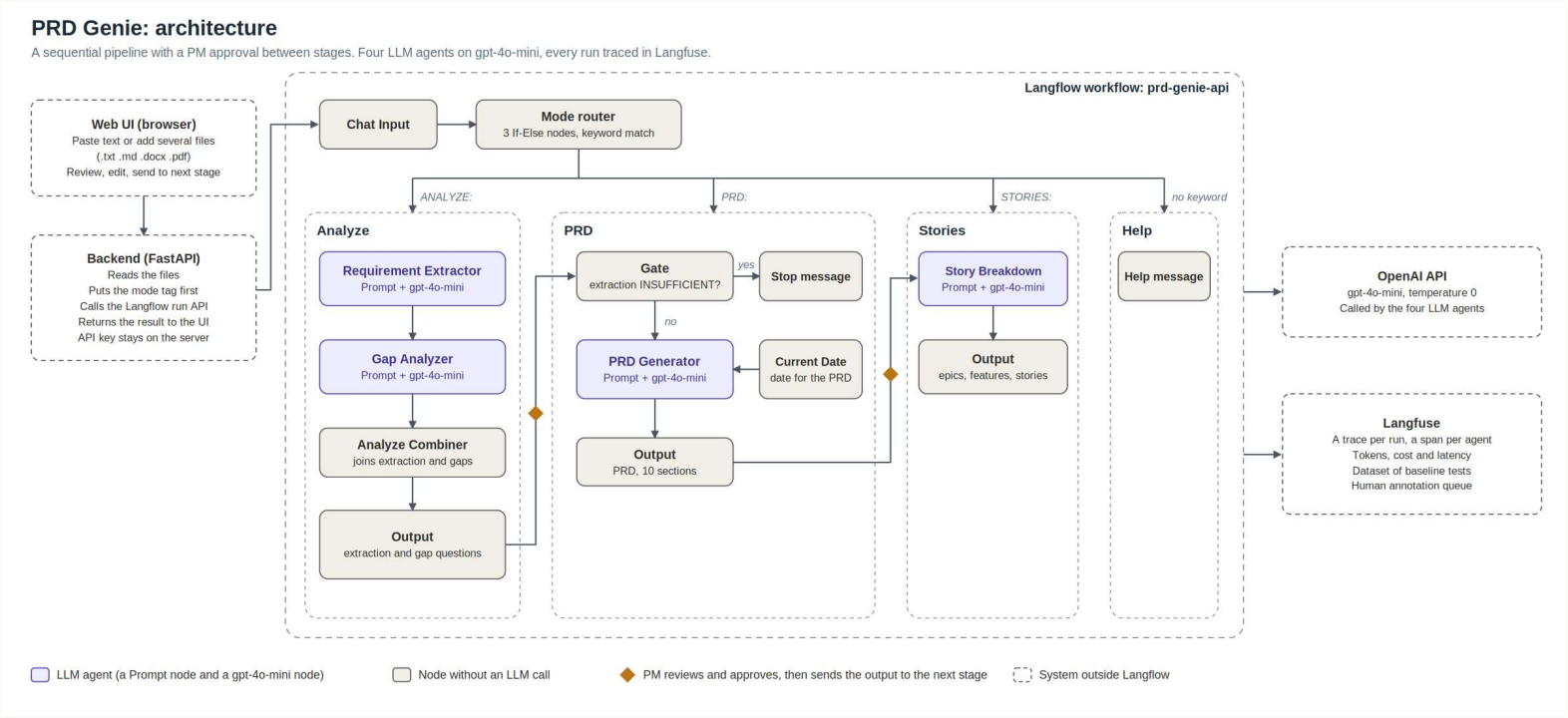
Fills a fixed 10-section template, using only the approved extraction.

Stories

Writes epics, features and stories that trace back to the PRD.

- The PM reviews and approves each result before the next stage runs.

How it fits together



Live demo

1. **Vague notes are flagged, not filled in**
   The agent asks for users, metrics and the output format

2. **Detailed requirements run through all three stages**
   One chain, with the PM approving between stages

3. **An empty input is stopped by the gate**
   No PRD is built from nothing

Then the Langfuse trace: tokens, cost and latency for each agent

Results on the baseline set

11 of 12

tests meet their criteria

17

stage runs recorded, with every output

1 of 17

runs with a small unsupported detail

What worked

Vague input flagged (T2, T5, T9). Exact numbers kept (T7). An explicit conflict found (T6). Criteria copied verbatim (T4).

What failed

T3: an implicit trade-off between two statements is not detected, though explicit disagreements are.

Every item must trace back to the source

Exact quotes

Every requirement carries the words that were said.

No guessing

Missing fields read “Not specified in source”.

A gate

An extraction marked INSUFFICIENT never becomes a PRD.

Labelled drafts

Priority and benefits come from the source, or stay a visible placeholder.

Each failure became a rule, then a re-test

Failure found	Fix	Result after the re-test
The stories invented defaults and message texts	A stricter prompt: criteria in quotes, no drafted rules	T4, T8 and T12 pass with no invented specifics
An empty input gave no useful question (T9)	A rule: INSUFFICIENT must name what is missing	The Gap agent now asks for it
An invented risk, priority and deadline	A rule: copy only words that appear in the input	Gone in the final runs
Stories invented benefits and priorities	A placeholder benefit; priority only from the PRD	No invented benefits

One passing run is not proof, so the baseline set is re-run after every prompt change

A full run costs about $0.002

$0.0022

for one chain: Analyze, PRD and Stories

36 s

from notes to stories

$0.17

per PM per month, at 3 documents a day (an assumption)

Stories is 55% of the cost and 53% of the time.

Output tokens are 25% of the tokens but 58% of the cost.

Measured in Langfuse on one chain; the monthly figure uses an assumed rate

Known limits and what comes next

Known limits

- An implicit trade-off is missed (T3)
- Output varies between runs
- Stories are thin by design

Next

- A questions-first step: the PM answers before stories are written
- Ask which one is meant (which API, which competitor)
- A pilot with 3 to 5 PMs to measure time saved

Code, prompts, results and documents: [GitHub link]